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Intermetallic phase evolution of 7050 aluminum alloy during homogenization

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ABSTRACT

Intermetallic phase evolution of 7050 aluminum alloy during homogenization was investigated in detail by optical microscopy, scanning electron microscopy, energy dispersive spectrometry, differential scanning calorimetry, electron probe micro-analysis and X-ray diffraction methods. The results show that severe dendritic segregation exists in as-cast alloy. The dissolvable intermetallic phases in as-cast alloy consist of equilibrium η ($MgZn_2$) phase, Cu and Mg enriched non-equilibrium aluminides and Cu enriched non-equilibrium aluminides. During homogenization, Cu and Mg enriched non-equilibrium aluminides, Cu enriched non-equilibrium aluminides and η ($MgZn_2$) phase gradually dissolve into matrix. Equilibrium S (Al_2CuMg) intermetallic phase nucleates and grows along the grain boundaries, and it disappears completely after multi-step homogenization. The proper homogenization processing is 400 °C × 10 h + step470 °C × 24 h + step485 °C × 4 h, which is consistent with the results of homogenizing kinetic analysis.

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1. Introduction

Considerable commercial interest has been shown in the development of high strength Al–Zn–Mg–Cu base 7XXX series aluminum alloys for aerospace constructional uses, due to a superior combination of high specific strength, fracture toughness and stress corrosion cracking resistance [1,2]. Usually the residual coarse intermetallic phases (>1 μm) will deteriorate the toughness and fatigue performance [3–5]. In order to get good working qualities and service performance, a homogenization heat treatment is carried out directly after casting before processing, and it plays a key role in removing the microsegregation and dissolving large soluble non-equilibrium intermetallic phases formed during rapid solidification processing [6–8].

Existing literature indicates that below solidus temperature, four major intermetallic phases, η, T, S and θ can occur during rapid solidification processing in commercial Al–Zn–Mg–Cu alloys [9–11]. Moreover, the type and intrinsic character of intermetallic phases will differ in different alloys or in one alloy at different solidified and homogenized conditions. Under the present homogenization treatment, all the dissolvable intermetallic phases present in the as-cast Al–Zn–Mg–Cu microstructures dissolve into matrix completely, except for S (Al_2CuMg) phase [12,13]. Furthermore, it remains more residuum in the aluminum alloy with more copper additives, which affects the subsequent processing and deteriorates service performance. Due to the higher content of alloying elements in 7050 alloy, it is more important to investigate its evolution of intermetallic phases and to develop a new homogenization process for actual industrial applications. Besides, the homogenization kinetic equation of 7050 alloy was also established successfully.

2. Material and methods

The semi-continuous 7050 ingots were provided by Northeast Light Alloy Co., Ltd. The composition of the studied ingots is Al-5.87 Zn-2.07 Mg-2.28 Cu-0.11 Zr-0.04 Si-0.08 Fe-0.10 Mn-0.04 Cr-0.05 Ni-0.06 Ti (wt.%). Ten ingot homogenization conditions were used as below. Condition A: 400 °C × 10 h; Condition B: 470 °C × 24 h; Condition C: 400 °C × 10 h + step470 °C × 24 h; Condition D: 400 °C × 10 h + step470 °C × 24 h + step485 °C × 10 min; Condition E: 400 °C × 10 h + step470 °C × 24 h + step485 °C × 30 min; Condition F: 400 °C × 10 h + step470 °C × 24 h + step485 °C × 1 h; Condition G: 400 °C × 10 h + step470 °C × 24 h + step485 °C × 4 h; Condition H: 400 °C × 10 h + step470 °C × 24 h + step485 °C × 9 h; Condition I: 400 °C × 10 h + step470 °C × 24 h + step485 °C × 12 h; Condition J: 400 °C × 10 h + step470 °C × 24 h + step485 °C × 24 h. The present industrial homogenizing process is 400 °C × 10 h + step470 °C × 24 h. The purpose of the pretreatment at 400 °C for 10 h is to obtain fine Al_3Zr particles. Homogenizing at

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470 °C for 24 h is to eliminate low-melting intermetallic phases [14–17]. Owing to many existing reports about Al_3Zr particles [18–22], our research no longer focuses on their behaviors. In addition, we designed a third step homogenization at 485 °C, trying to eliminate S (Al_2CuMg) intermetallic phases effectively.

Differential scanning calorimetry (DSC) analyses were conducted using a SDT-Q600 differential scanning calorimeter at a constant heating rate of 10 °C/min from 25 to 650 °C, and the standard sample is Al_2O_3. A combination of optical microscopy (OM), scanning electron microscopy (SEM) and electron probe micro-analysis (EPMA) was carried out to characterize as-cast and homogenized microstructures. SEM, in the backscattered electron imaging mode, was carried out on the SEM attached to the EPMA instrument, operating at 20 kV. EPMA was performed on a JEOL JXA-8230 Electron Probe Microanalyzer operating at 20 kV. A quantitative X-ray wavelength dispersive spectroscopy (WDS) system attached to the EPMA instrument was used to analyze the chemistry of the intermetallic phases present in as-cast and homogenized microstructures. For each type of phase, at least three particles from different locations were analyzed and the average values of such analyses were reported here. The map and line scanning analyses were performed on an energy dispersive X-ray (EDX) measurement attached to a Quanta MK2-200 scanning electron microscope (SEM), operating at 20 kV, to observe the distribution of the elements in the as-cast and homogenized materials. The analyses were carried out on polished but unetched samples. X-ray diffraction studies (XRD) were undertaken in order to identify the alloy phases present in as-cast and homogenized materials. X-ray diffraction studies were performed on a D/max 2500PC diffractometer with $CuK\alpha_1$ radiation.

3. Results

3.1. Characterization of as-cast microstructure

Fig. 1 shows the microstructure of as-cast alloy. It is a typical as-cast eutectic structure, exhibiting serious dendritic segregation, which requires further heat treatment process to eliminate it (Fig. 1(a)). The average grain size of as-cast alloy is about 121 μm. The lamellar eutectic structures can be observed in Fig. 1(b). There are two kinds of intermetallic phases in as-cast microstructure. Quantitative WDS analyses show that the chemical composition of white intermetallic phase (as arrowed in Fig. 1(b) A) follows the range: 18.76–21.07 at.% Cu, 24.23–27.81 at.% Mg, 3.51–4.04 at.% Zn and the balance being aluminum. The chemical composition of grey intermetallic phase (as arrowed in Fig. 1(b) B) follows the range: 24.01–26.82 at.% Cu, 3.93–4.28 at.% Mg, 0.61–0.65Zn at.% and the balance being aluminum. Although the chemical composition of the white intermetallic phase is close to that of S (Al_2CuMg) phase, the Cu content of this phase is less than that of S (Al_2CuMg) phase. Similarly, the chemical composition of the grey intermetallic phase is close to that of θ (Al_2Cu) phase, but the Cu content of this phase is less than that of θ (Al_2Cu) phase. Besides, the white intermetallic phase dissolves small amounts of zinc in it, and the grey intermetallic phase dissolves small amounts of magnesium in it. There are some differences between those two phases and S (Al_2CuMg) and θ (Al_2Cu) phase. Therefore, we cannot define those two intermetallic phases as S (Al_2CuMg) or θ (Al_2Cu) simply. According to the existing references, there are no similar reports. As a result, we defined the white intermetallic phase as Cu and Mg enriched non-equilibrium aluminides, and defined the grey intermetallic phase as Cu enriched non-equilibrium aluminides, respectively.

Fig. 2 shows the scanning electron microstructure and the main elements Zn, Mg and Cu distribution in as-cast alloy. The main elements Zn, Mg and Cu are largely enriched in grain boundaries, and the concentration of the elements decreases from grain boundary to inside. Therefore, a homogenization treatment is required to removing severe microsegregation in as-cast 7050 alloy.

3.2. DSC analysis

DSC curves of as-cast and homogenized 7050 aluminum alloy are shown in Fig. 3. The area under a given DSC peak relates to the phase volume fraction, calculated qualitatively by its enthalpy value. The peak temperature of a given reaction is dependent on the phase composition under the same condition [23]. The temperatures and enthalpy values, corresponding to different endothermic dissolution peaks obtained from DSC curves in Fig. 3, are shown in Table 1.

Three endothermic peaks, sited at 475.2 °C, 486.7 °C and 633.5 °C respectively, are observed in as-cast alloy. After homogenizing at 400 °C for 10 h, the first endothermic peak, sited at 475.2 °C, still exists, but the enthalpy value corresponding to this peak decreases. Therefore, it can be inferred that the intermetallic phase, melt at 475.2 °C, still exists but partially dissolves into matrix after homogenizing at 400 °C for 10 h. This intermetallic phase totally dissolves into matrix under homogenization conditions B and C. Besides, seen from Table 1, the pretreatment has little effect on the evolution of intermetallic phase. The second endothermic peak is present at different temperatures under different homogenization conditions. It disappears under homogenization condition G. The enthalpy value corresponding to this endothermic peak gradually decreases for the duration of the third step homogenization. As a result, the remnant intermetallic phase corresponding to the second endothermic peak gradually dissolves for the duration of the third step homogenization. This intermetallic

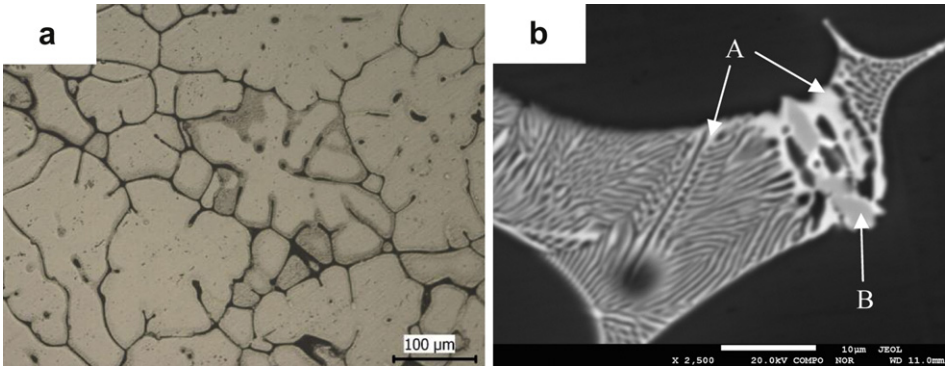

Fig. 1. Microstructures of the as-cast alloy: (a) optical microstructure; (b) backscattered electron image.

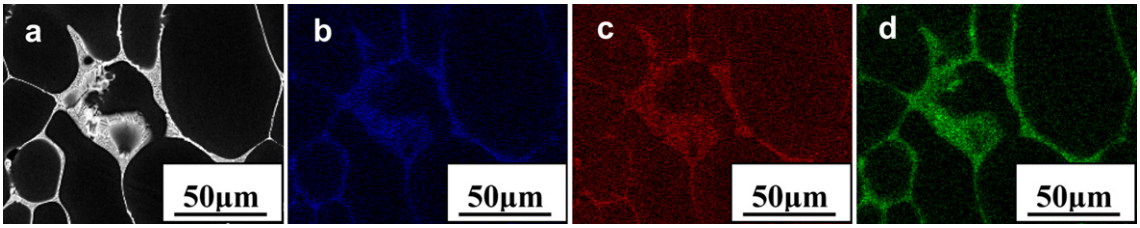

Fig. 2. Scanning electron microstructure and the main elements distribution in as-cast alloy: (a) backscattered electron image; (b) Zn; (c) Mg; (d) Cu.

phase finally dissolves into matrix completely under homogenization condition G. The type of the intermetallic phase corresponding to the second endothermic peak is different between as-cast and homogenized alloy because the temperature of the second endothermic peak in homogenized alloy is higher than that in the as-cast alloy. The third endothermic peak sited at 633.5 °C is ascribed to the melting point for alloy. Therefore, considering the first-step homogenization for obtaining Al_3Zr dispersoid particles, the upper limit temperatures of the second step and the third step homogenization are 475.2 °C and 492 °C, respectively.

3.3. Optical microstructure

The optical microstructure of homogenized alloy is shown in Fig. 4. Large non-equilibrium intermetallic phases distribute in grains and along grain boundaries after pretreatment at 400 °C for 10 h (Fig. 4(a)). The volume fraction of the residual phases and dendritic-network structure is significantly reduced after the second step or the third step homogenization, at the same time, the grain boundaries become thinner (Fig. 4(b) and (c)). There is no obvious increase of the average grain sizes in 7050 alloy after different homogenization treatment. From Fig. 4(c) and (d), it can be seen that there is no distinct change of optical microstructure by prolonging the third step homogenization time, and the over-burnt phenomenon is not observed.

3.4. Evolution of intermetallic phases during homogenization

Backscattered images of homogenized alloy are shown in Fig. 5. The composition of the intermetallic phases present in Fig. 5, obtained from Quantitative X-ray WDS analysis, is shown in Table 2.

After homogenizing at 400 °C for 10 h, Cu enriched non-equilibrium aluminides present in as-cast microstructure disappears. At the same time, a new grey intermetallic phase forms. According to the quantitative X-ray WDS analysis of this grey phase, the ratio of Cu to Mg is close to 1:1. Moreover, there is no other element in the phase except for Al matrix. Therefore, we consider that grey intermetallic phase is S (Al_2CuMg) phase. Seen from Fig. 5(a), the S (Al_2CuMg) phase forms and grows along the boundary between Cu and Mg enriched non-equilibrium aluminides and the matrix. S (Al_2CuMg) phase gradually dissolves into matrix during homogenization. Only small residual S (Al_2CuMg) phases coarsen at the grain boundaries under homogenization condition C. At the same time, Cu and Mg enriched non-equilibrium aluminides disappears (Fig. 5(b)). From the quantitative X-ray WDS analysis, with the duration of the third step homogenization, the Cu and Mg content of S (Al_2CuMg) phase is closer to 1:1. During homogenization treatment, S (Al_2CuMg) phase gradually disappears and finally dissolves into matrix completely under condition G (Fig. 5(c and d)). Only an irregular phase exists under conditions G and I (Fig. 5(e and f)). From the quantitative X-ray WDS analysis of this irregular phase, the ratio of Cu to Fe is 2:1. Moreover, there is no other element in this phase except for Al matrix. Therefore, we consider that grey intermetallic phase is Al_7Cu_2Fe phase. Al_7Cu_2Fe phase (arrowed as F and G) exhibits no change with the increase of the third step homogenization time (Fig. 5(e) and (f)). Because Al_7Cu_2Fe phase is an indissolvable impurity intermetallic phase, it can't be eliminated by homogenization treatment.

According to the results of thermal analysis and quantitative X-ray WDS analysis, it can be concluded that the first endothermic peak, sited at 475.2 °C, is corresponding to the melting of Cu and Mg enriched non-equilibrium aluminides, and the second endothermic

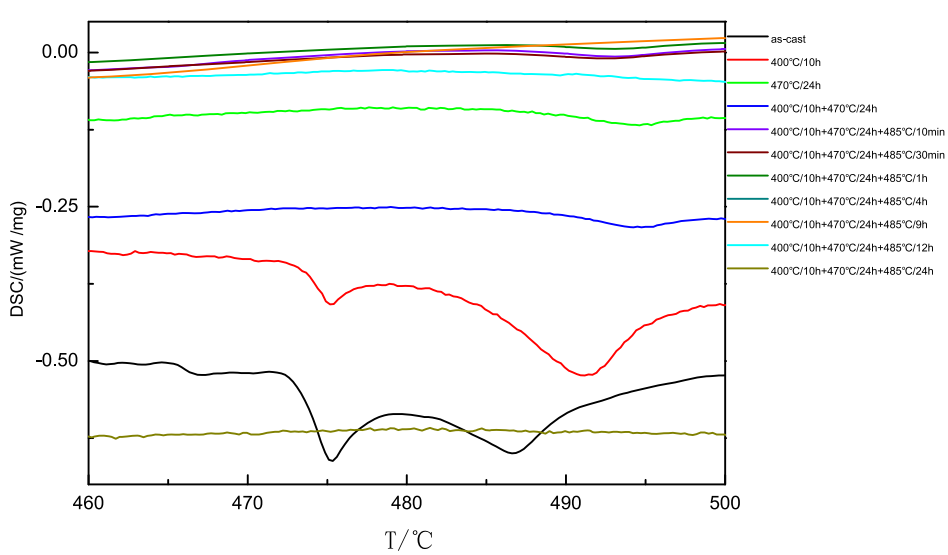

Fig. 3. DSC curves of as-cast and homogenized alloy.

Table 1
The temperatures and enthalpy values corresponding to endothermic peaks of thermal analyses in Fig. 3.

Homogenization condition	The first peak		The second peak		The third peak
	T (°C)	ΔH (J/g)	T (°C)	ΔH (J/g)	T (°C)
As-cast	475.2	−1.962	486.7	−3.168	633.5
A: 400 °C/10 h	475.2	−0.802	493.1	−6.093	
B: 470 °C/24 h	disappeared	0	495.3	−0.691	
C: 400 °C/10 h+470 °C/24 h	disappeared	0	494.8	−0.624	
D: 400 °C/10 h+470 °C/24 h+485 °C/10 min	disappeared	0	493.1	−0.502	
E: 400 °C/10 h+470 °C/24 h+485 °C/30 min	disappeared	0	492.5	−0.429	
F: 400 °C/10 h+470 °C/24 h+485 °C/1 h	disappeared	0	492.9	−0.338	
G: 400 °C/10 h+470 °C/24 h+485 °C/4 h	disappeared	0	disappeared	0	
H: 400 °C/10 h+470 °C/24 h+485 °C/9 h	disappeared	0	disappeared	0	
I: 400 °C/10 h+470 °C/24 h+485 °C/12 h	disappeared	0	disappeared	0	
J: 400 °C/10 h+470 °C/24 h+485 °C/24 h	disappeared	0	disappeared	0	

peak at 486.7 °C in as-cast alloy is ascribed to the melting of Cu enriched non-equilibrium aluminides. Moreover, the second endothermic peaks in homogenized alloys are all the melting of S (Al_2CuMg) phase. The difference of the melting temperature of S (Al_2CuMg) phase may result from the changes of Cu and Mg content.

3.5. X-ray diffraction analysis

X-ray diffraction patterns are shown in Fig. 6. The phases in as-cast alloy consist of α (Al) and η ($MgZn_2$). After pretreatment, η ($MgZn_2$) phase decreases, and S (Al_2CuMg) phase forms. η ($MgZn_2$) phase disappears under homogenization condition C. There is no obvious diffraction peak in X-ray diffraction patterns except for α (Al) under homogenization conditions C, G and I. In our analysis, Cu and Mg enriched non-equilibrium aluminides, Cu enriched non-equilibrium aluminides and Al_7Cu_2Fe phase are not detected by X-ray diffraction because of their low contents and η phase is not observed in scanning microstructures due to its small size.

Therefore, the dissolvable intermetallic phases in as-cast alloy consist of equilibrium η ($MgZn_2$), Cu and Mg enriched non-equilibrium aluminides and Cu enriched non-equilibrium aluminides. After homogenization at 400 °C for 10 h, Cu enriched non-equilibrium aluminides disappears, and S (Al_2CuMg) phase forms. Under industrial homogenization condition, Cu and Mg enriched non-equilibrium aluminides and η ($MgZn_2$) phase disappear, and only small S (Al_2CuMg) phase exists. Because no soluble intermetallic phase exists under condition G, the proper homogenization processing for the studied alloy is 400 °C × 10 h + step470 °C × 24 h + step485 °C × 4 h.

3.6. Line scanning analysis

Line scanning traces of the as-cast and homogenized alloy are shown in Fig. 7. There is almost no obvious change of the main elements segregation between the as-cast alloy and the alloy homogenized at 400 °C for 10 h. The segregation of Zn is eliminated under condition C. The Mg element distribution is homogeneous

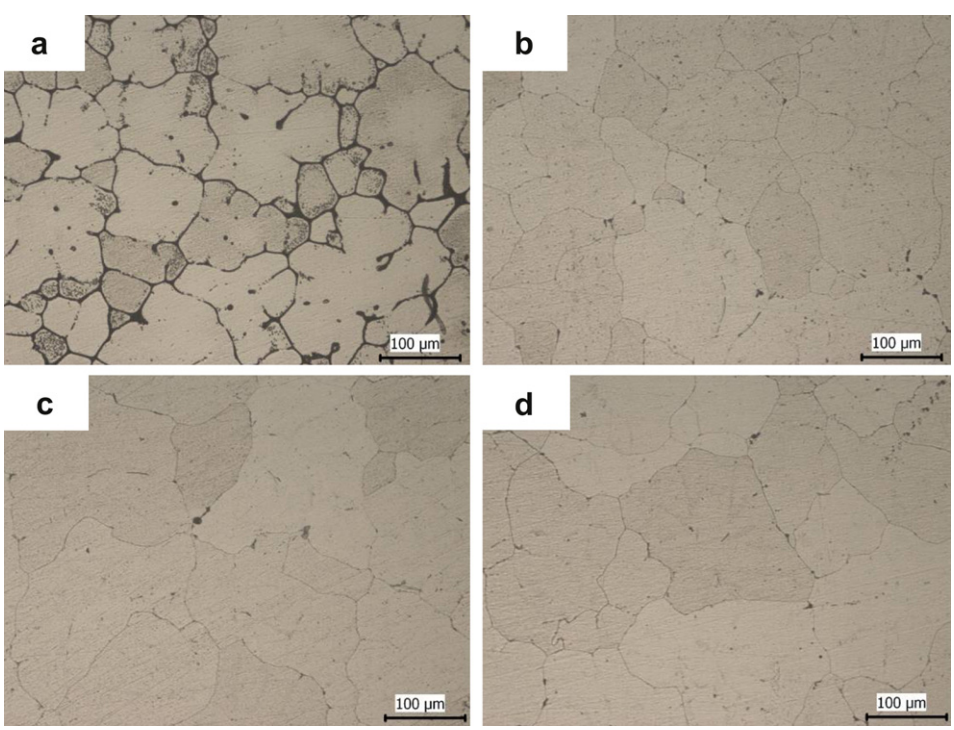

Fig. 4. Optical micrographs of homogenized alloy: (a) 400 °C × 10 h; (b) 400 °C × 10 h + step470 °C × 24 h; (c) 400 °C × 10 h + step470 °C × 24 h + step485 °C × 4 h; (d) 400 °C × 10 h + step470 °C × 24 h + step485 °C × 12 h.

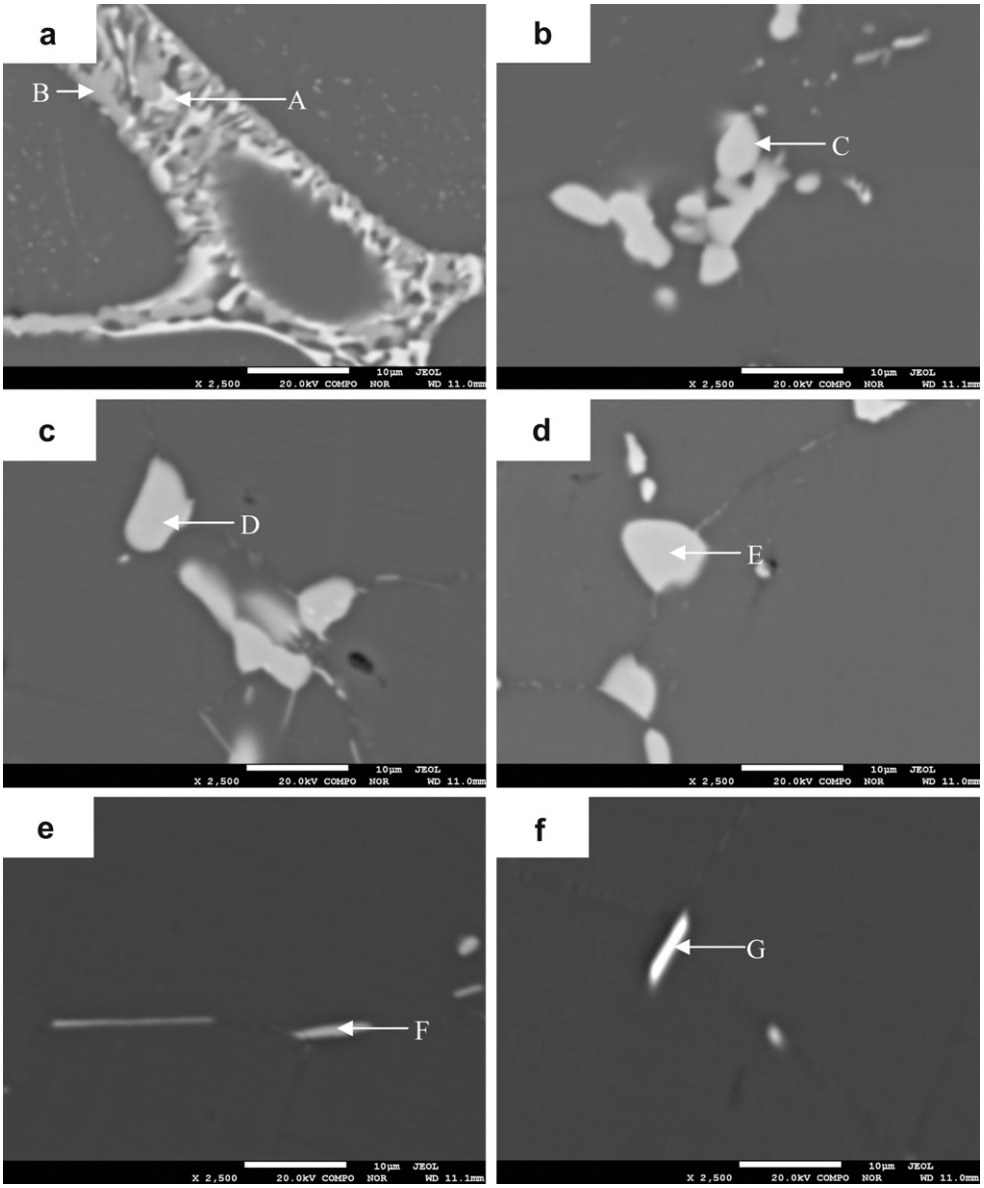

Fig. 5. Backscattered electron images of homogenized alloy: (a) 400 °C × 10 h; (b) 400 °C × 10 h + step470 °C × 24 h; (c) 400 °C × 10 h + step470 °C × 24 h + step485 °C × 10 min; (d) 400 °C × 10 h + step470 °C × 24 h + step485 °C × 1 h; (e) 400 °C × 10 h + step470 °C × 24 h + step485 °C × 4 h; (f) 400 °C × 10 h + step470 °C × 24 h + step485 °C × 12 h.

from the grain boundary to inside and only small segregation of Cu and Fe exists under condition I. Therefore, we can conclude that, the diffusion velocity of main elements is Zn > Mg > Cu, and the homogenization temperature to eliminate the main elements segregation is Cu > Mg > Zn.

3.7. Homogenization kinetic analysis

It can be seen that the distribution of the main elements along interdendritic region varies periodically from Fig. 7. Therefore, the studies of diffusion law along interdendritic region are

Table 2
Chemical composition of intermetallic phases in Fig. 5 (in at.%).

Phase	A	B	C	D	E	F	G
Si	0.01	0.01	0.00	0.03	0.01	0.00	0.00
Al	48.93	61.64	54.93	53.56	51.60	72.61	75.01
Mg	28.04	19.12	23.16	23.12	24.36	0.17	0.62
Ti	0.01	0.00	0.00	0.00	0.01	0.20	0.15
Zn	4.72	0.58	0.22	0.26	0.20	0.11	0.10
Cu	18.21	18.60	21.68	22.99	23.75	17.75	15.89
Fe	0.05	0.04	0.02	0.02	0.00	8.82	8.00
Mn	0.01	0.03	0.00	0.01	0.00	0.03	0.01
Ni	0.00	0.08	0.00	0.00	0.01	0.31	0.22
Zr	0.02	0.00	0.00	0.02	0.05	0.00	0.00
Closest phase	Cu and Mg enriched	Al_2CuMg	Al_2CuMg	Al_2CuMg	Al_2CuMg	Al_7Cu_2Fe	Al_7Cu_2Fe

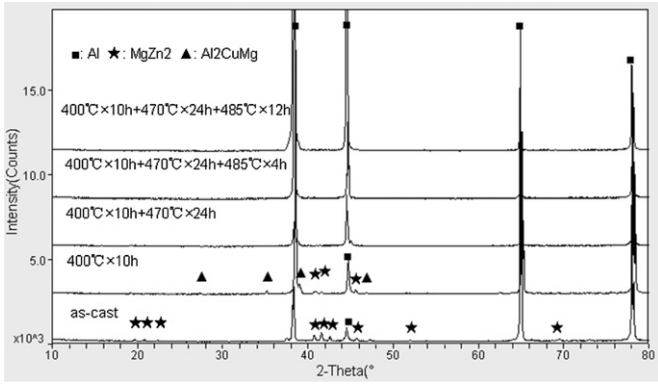

Fig. 6. X-ray diffraction patterns.

important to the investigations of elements distribution during homogenization.

According to Shewman [5,24], the initial concentration of the elements along the interdendritic region can be approached by Fourier series components in a cosine function:

$$C(x) = \bar{C} + A_0 \cos\frac{2\pi x}{L} \tag{1}$$

where $\bar{C}$ is the average concentration of the element, L is the interdendritic spacing and A_0 is the initial amplitude of the composition segregation, expressed as

$$A_0 = \frac{1}{2}(C_{max} - C_{min}) = \frac{1}{2}\Delta C_0 \tag{2}$$

According to the second Fick's law [25–29] and the boundary conditions, $A(t)$ is given as

$$A(t) = A_0 \exp\left(-\frac{4\pi^2}{L^2}Dt\right) \tag{3}$$

By substituting Eq. (1) into Eq. (4), the equation can be rewritten as

$$A(t) = A_0 \exp\left[-\frac{4\pi^2 D_0 t}{L^2}\exp\left(-\frac{Q}{RT}\right)\right] \tag{4}$$

Assuming the element distribution is homogeneous when the composition segregation amplitude is reduced to 1%, i.e.:

$$\frac{A(t)}{A_0} = \frac{1}{100} \tag{5}$$

then

$$\exp\left[-\frac{4\pi^2 D_0 t}{L^2}\exp\left(-\frac{Q}{RT}\right)\right] = \frac{1}{100} \tag{6}$$

By taking natural logarithms of both sides, Eq. (6) can be rewritten as

$$\frac{1}{T} = \frac{R}{Q}\ln\left(\frac{4\pi^2 D_0 t}{4.6L^2}\right) \tag{7}$$

Assuming $A = R/Q$ and $B = 4.6/4\pi^2 D_0$, we can obtain:

$$\frac{1}{T} = A\ln\left(\frac{t}{BL^2}\right) \tag{8}$$

Eq. (8) is the homogenization kinetic equation. If the parameters of as-cast microstructure are given, the homogenization kinetic curves can be obtained. From the results of Section 3.6 and previous reference [30], it can be concluded that the diffusion coefficient of Cu is much lower than Mg, Zn at the same temperature. Therefore,

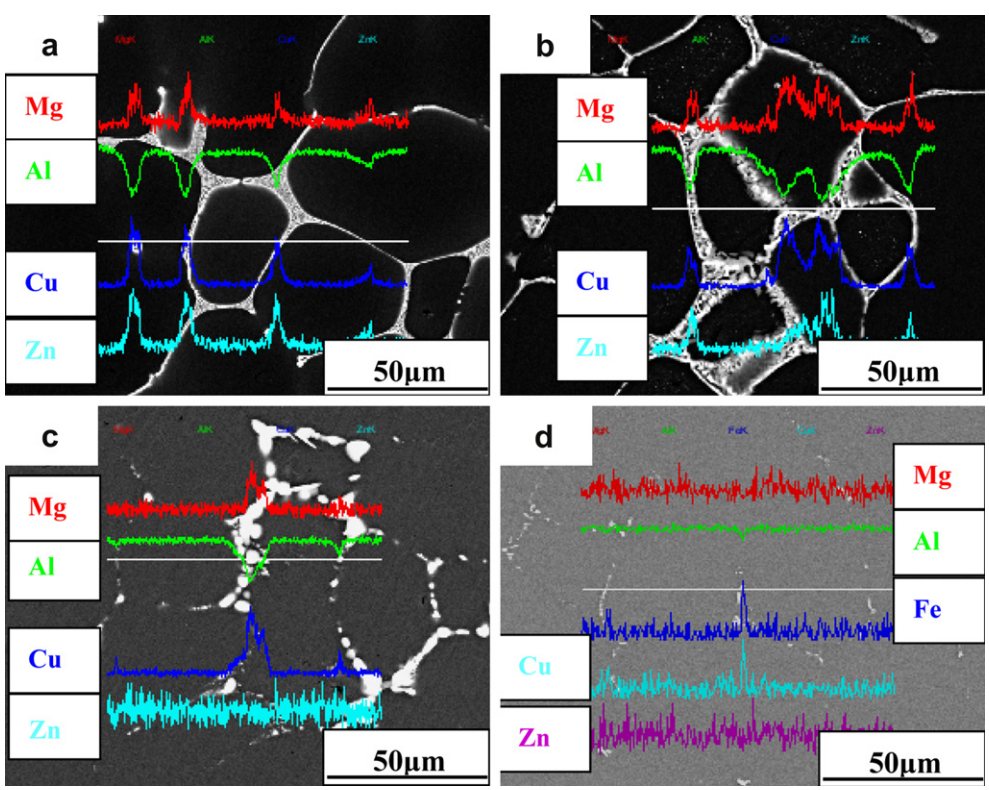

Fig. 7. Line scanning traces of the alloy under different conditions: (a) as-cast; (b) 400 °C × 10 h; (c) 400 °C × 10 h + step470 °C × 24 h; (d) 400 °C × 10 h + step470 °C × 24 h + step485 °C × 12 h.

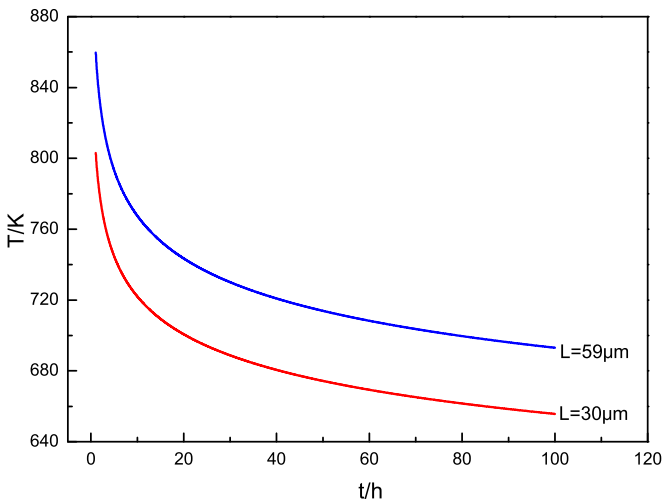

Fig. 8. Curves of homogenization kinetics.

the homogenization process is believed to be controlled by the diffusion of Cu. By substituting of $D_0(\mathrm{Cu}) = 0.084\ \mathrm{cm^2/s}$, $Q(\mathrm{Cu}) = 136.8$ kJ/mol and $R = 8.31$ J/(mol K) into Eq. (8), the homogenization kinetic curves of 7050 alloy for different interdendritic spacings can be obtained.

By substituting the average interdendritic spacing L into Eq. (8), the suitable homogenizing parameters are obtained. The average interdendritic spacing of the studied ingot after the first step and the second step homogenization treatment is 59 μm and 30 μm respectively, obtained from quantitative metallographic analyses. According to the homogenization kinetic curves as shown in Fig. 8, at the optimized temperatures of 470 °C and 485 °C, the corresponding homogenization time is 20.3 h and 3.4 h, respectively. This is in good accordance with the experimental results.

4. Conclusions

(1) Serious dendritic segregation exists in the as-cast alloy. The main elements Zn, Mg and Cu are largely enriched in the grain boundaries. The dissolvable interdendritic phases in as-cast alloy contain equilibrium η ($MgZn_2$), Cu and Mg enriched non-equilibrium aluminides and Cu enriched non-equilibrium aluminides.

(2) The evolution for interdendritic phases during homogenization is as follows. Cu enriched non-equilibrium aluminides dissolve into matrix. S (Al_2CuMg) phase forms and grows along the boundary between Cu and Mg enriched non-equilibrium aluminides and the matrix. Cu and Mg enriched non-equilibrium aluminides and η ($MgZn_2$) phase disappear. S (Al_2CuMg) phase dissolves into matrix completely after multi-step homogenization. Besides, the morphology of insoluble Al_7Cu_2Fe interdendritic phase is not regular and it exhibits no change during homogenization.

(3) The diffusion velocity of main elements is Zn > Mg > Cu, and the homogenization temperature to eliminate the main elements segregation is Cu > Mg > Zn.

(4) The proper homogenization processing is 400 °C × 10 h + step470 °C × 24 h + step485 °C × 4 h, which is consistent with the results of homogenizing kinetic analysis.

Acknowledgements

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